

1 SAFETY

1.1	Summary	1-1
1.2	Warning Indications: Types and Definitions	1-2
1.3	General Precautions to be Observed	1-3
1.4	Special Safety Precautions	1-5
1.5	Warning Labels	1-6
1.5.1	Warning Label Handling	1-6
1.5.2	Warning Label Location	1-7
1.6	Drive Power Shutdown and Indication	1-8
1.7	Main Components	1-9
1.8	Configuration Diagram	1-11

1 SAFETY

1.1 Summary

The information in this chapter is included to instruct persons who own, install, operate, service or inspect this equipment on warning symbols, precautions which must be observed, and explanation of warning labels attached to the machine.



- **Before attempting to perform any operation, maintenance or inspection of this equipment, it is imperative to read and understand the instructions in this manual and to carefully observe all safety precautions and warnings contained herein.**
 - **If there are any unclear points or questions concerning the information contained in this manual, please contact Ishida installation personnel or your nearest Ishida Service representative before proceeding.**
-

1.2 Warning Indications: Types and Definitions

The warning indications contained in this manual, as well the indications on the labels attached to the machine, are ranked into three categories according to the level of hazard involved. It is important that these indications be fully understood and complied with.

INDICATION	EXPLANATION
 DANGER	If this hazard is not avoided, death or serious injury will probably result. This indicates a clear and immediate danger, and extreme caution must be exercised to prevent a mishap.
 WARNING	If this hazard is not avoided, there is a possibility of death or injury resulting.
 CAUTION	If this hazard is not avoided, there is a possibility that light or moderate injury may result. It may also indicate that a possibility of damage to equipment exists.
 NOTE	Used to emphasize or clarify an important point in the manual.

1.3 General Precautions to be Observed

This section describes the general safety precautions which must be observed when handling this equipment.

DANGER

- All electrical work for the installation site must be performed by licensed electrical contractors. Otherwise, electrical shock or equipment malfunction may result.
- Do not put hands through the gap at the bottom of the cover on the equipment's front. There are rotating parts inside the cover which could injure hands.
- All maintenance and inspection work involving electrical components must be performed by qualified maintenance personnel.
Electrical shock or equipment malfunction may result if unqualified personnel are permitted to perform maintenance or inspection of electrical components.
- Before performing any maintenance or inspection work which is not specifically indicated in this manual, shut off the main power switch.
The person performing the maintenance should lock the main power switch and keep the key in his possession while performing the work.
Injury or electrical shock may result if the equipment is turned on by another person while maintenance work is being performed.
- Turn off the main power switch then wait at least 3 minutes before performing any maintenance or inspection work on the equipment. Residual charge may remain in the machine even after the power has been turned off.
Also, turn off the power of any upstream/downstream units which are connected or adjacent to the equipment; otherwise there is a danger of electrical shock.

WARNING

- Persons who operate, maintain or inspect the equipment must have received appropriate training, and have attained a sufficient level of skill before they start performing such work.
- Never touch any electrical switches or buttons with wet or damp hands. Electrical shock may occur when the equipment is not properly grounded or when there is electrical leakage.
- Personnel with long hair using this equipment should tie up their hair securely and all personnel must wear a cap or hat as well as clothes and shoes suitable for the production environment. Unbound long hair or inappropriate clothing may become caught in moving parts, and injuries may result.
- Before starting operation of the equipment, make sure that all covers are securely shut and fastened. Touching the moving part during production may result injuries.
- Keep away from moving and rotating components while the equipment is in operation. They could cause injury if body parts get caught up in them.

- Before cleaning the part which is not specifically indicated in this manual, shut off the main power switch.
The person performing the maintenance should lock the main power switch and keep the key in his possession while performing the work.
Injury or electrical shock may result if the equipment is turned on by another person while cleaning is being performed.
- If maintenance or inspection work is to be performed with the main power switch ON, clearly indicate this situation by posting a sign in the work area. This is to prevent other personnel from accidentally starting up the equipment.
- When performing maintenance and inspection work on the upper part of the packer, use a sturdy ladder or foothold to avoid a hazardous fall.

 CAUTION

- Do not run the equipment with any tools or other objects placed on top of the machine. If objects fall into moving parts, damage to the equipment may result.
- Use finger to press the operation panel of the RCU. Using a ball-point pen or other pointed object can damage the operation panel.

1.4 Special Safety Precautions

This section describes special safety precautions for this equipment which, in addition to the previously mentioned precautions, should be carefully observed.

WARNING

- Before opening the main guard and working on the parts behind it, make sure that the heater has thoroughly cooled. Otherwise the heater or its surrounding parts could cause burns when touched.

CAUTION

- Do not use the same power source for devices which may emit noise. Doing so may result in malfunction or damage.
- Do not install wiring which permits load-related power fluctuations of more than $\pm 10\%$. Overload may result in malfunction or damage.
- When cleaning the equipment, be sure to rinse only the areas specifically indicated as washable. Otherwise, damage or malfunction may result.
- Do not apply insecticide or other foreign substances inside the main body, terminal boxes, motor boxes, or other enclosed packer compartments. Doing so may cause machine malfunction or damage.

1.5 Warning Labels

Warning labels which indicate points requiring particular caution are attached to the packer at certain locations. Please take sufficient time to familiarize yourself thoroughly with the meanings and positions of these labels.

1.5.1 Warning Label Handling

- First verify that all the warning labels are clearly legible. If the label text or graphic is difficult to read, clean or replace the label.
- Clean labels with water and neutral cleanser.
Do not use organic solvents or gasoline.
- Labels must be replaced if they are damaged, peeled or illegible.
For this, please contact your local Ishida Service representative.

1.5.2 Warning Label Location

The diagram below shows the location of warning labels.

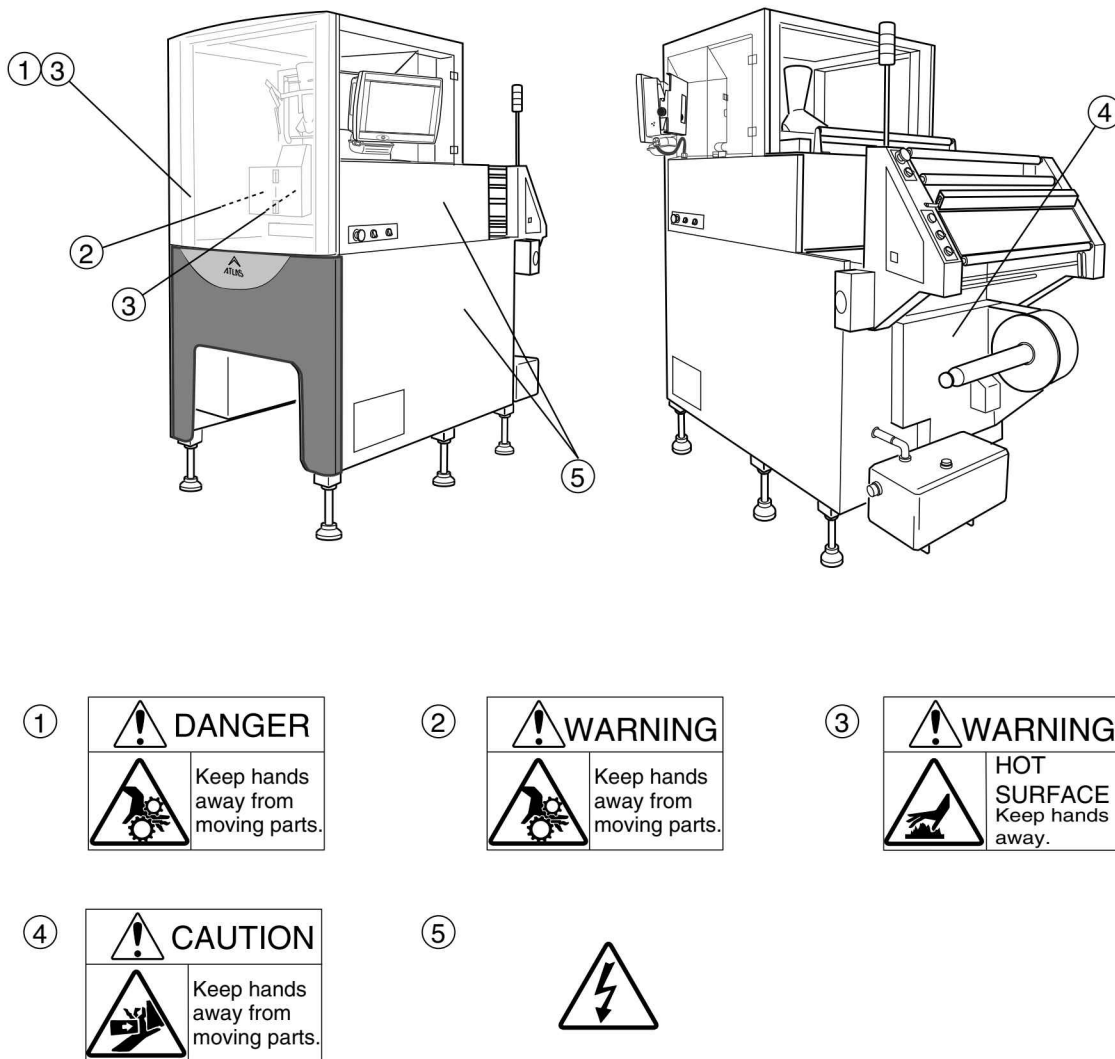


Fig. 1-1 Warning Label Location

1.6 Drive Power Shutdown and Indication

Before performing maintenance or inspection, electrical power should be shutdown to ensure the safety of personnel.

To prevent other personnel from starting operation while this work is being performed, follow the procedure described below.

- The main power switch should be shut off and locked.
- A tag clearly indicating that maintenance work is in progress should be prepared and posted on the power shutdown device.

Lock the main power switch and attach the accident prevention tag as shown below.

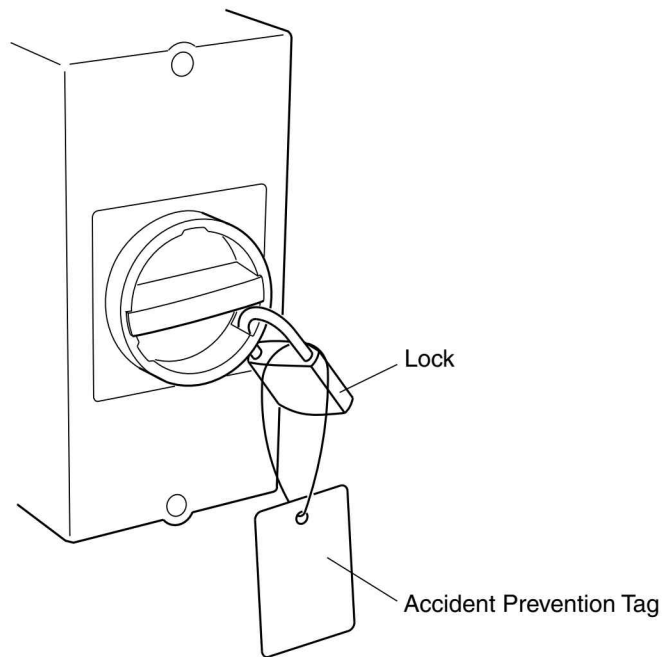


Fig. 1-2 Main Power Switch

1.7 Main Components

This section explains the main components and the functions.

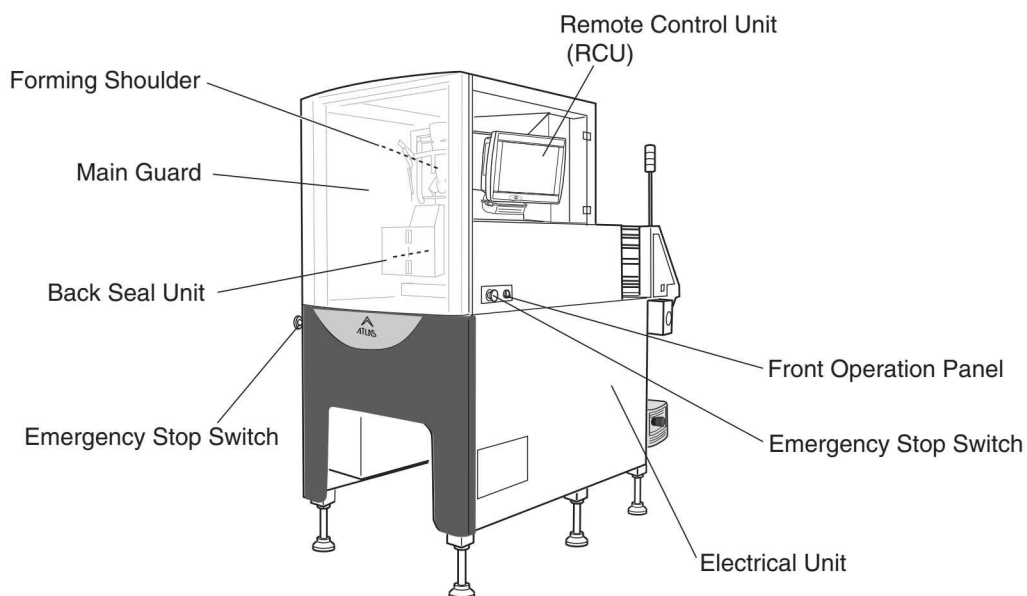


Fig. 1-3 External View (1)

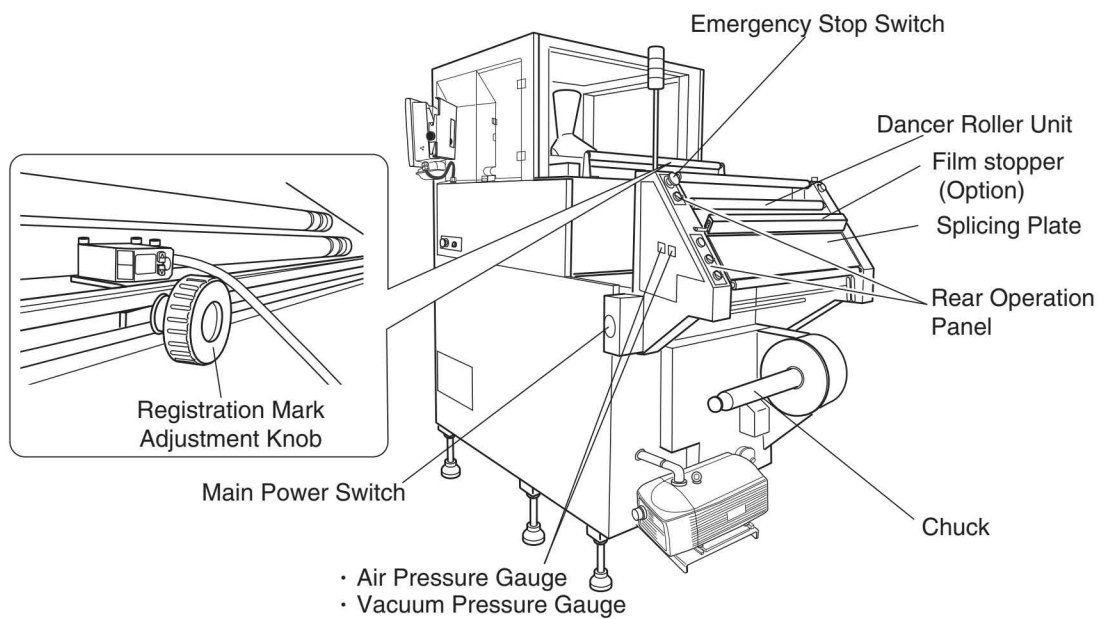


Fig. 1-4 External View (2)

Table 1-1 Explanation of Components

Unit	Functional Description
Emergency stop switch	Stops the equipment in an emergency.
Remote control unit (RCU)	Performs the settings and control operations necessary for operation.
Front operation panel	The emergency stop switch, Fill/Pause switch, Run/Lace switch, and knife switch are located here.
Main power switch	Supplies and shuts off the power to the equipment.
Air pressure gauge	Indicates the supply air pressure.
Rear operation panel	The restart switch, emergency stop switch, RUN/PAUSE switch, and Fill/Splice switch are located here.
Electrical unit	Houses the electrical control equipment.
Chuck	Clamps/unclamps the film reel.
Splicing plate	Retains the old film in place during film replacement.
Dancer roller unit	Absorbs the film's slack.
Printer (option)	Prints date, etc., on the film.
Registration mark adjustment knob	The registration mark sensor is integrated into this knob.
Forming shoulder	Forms the film into a tube.
Back seal unit	Seals the edges of the film-tube.
Horizontal seal unit	Seals the end of the film-tube, and cuts it.
Main guard	Transparent covering for the equipment's front